# Introduction to the Lecture

D. T. McGuiness, Ph.D
WS.2025
MCI

# Table of Contents

# Introduction

# Table of Contents

# Introduction

- The goal of this lecture is to give an introduction to the fascinating world of electric machines.
- We will start from understanding how they work to the types currenty used in commercial and industrial applications.
- The lecture will focus on mathematical and engineering concepts which will have direct use with real-life engineering applications.
- This lecture is a total of 3 SWS with a total of forty-five (45) UE.
- A unit (UE) is defined as 45 min lecture.

# Introduction

- Lecture materials and all possible supplements will be present in its Github Repo.
  - You can easily access the link to the web-page from here.

> Github is chosen for easy access to material management and CI/CD capabilities and allowing hosting websites.

- In the lecture content is also distributed as a WebBook which can be accessed from the Repo website.

# Introduction

- The student should have a good background in circuit analysis and familiarity with integral/differential calculus.
- After completion, the student will have a good grasp of electrical machines from their construction, how they are designed, what types out there and what applications are possible.

| Requirements | Taught Lecture | Code | Degree | Outcome |
|---:|---|---|---|---|
| Vector Calculus | Mathematik 1 | MAT | B.Sc | Python Programming |
| Differential Calculus | Mathematik 2 | MAT | B.Sc | Electric Drive Fundamentals |
| Linear Algebra | Mathematik 1 | MAT | B.Sc | Magnetic Circutis |
| Physics | Physik | PHY | | Electric Motor Analysis |
| - | | | | - |

**Table 1:** Distribution of materials across the semester.

# Introduction

| Description | Value |
|---:|---|
| Official Name | Antriebstechnik |
| Lecture Code | ANT |
| Module Code | MECH-B-5-ANT-ANT-ILV |
| Lecture Name | Drive Technology |
| Semester | 5 |
| Season | WS |
| Lecturer | Daniel T. McGuiness, Ph.D |
| Module Responsible | DaM |
| Software | Python, Simulink |
| SWS Total | 3 |
| UE Total | 45 |
| ECTS | 5 |
| Working Language | English |

# Introduction

- The lecture will have a single personal assignment comprising of a set list of questions and a final exam comprising of all the topics covered in the lecture. Instructions will be given in the assignment.
- For the written exam you are allowed to write your personal equation reference paper, as long as it is a single sheet of A4, double sided and contains no exercise or solutions. Your reference sheet will be collected after the exam.

| Assignment Type | Value |
|---:|---|
| Personal Assignment | 20 |
| Final Exam | 30 |
| Lab Work | 50 |
| Sum | 100 |

# Introduction

| Title |
| --- |
| Advanced electric drives: analysis control and modeling using MATLAB/Simulink |
| Analysis of electric machinery and drive systems |
| Design of rotating electrical machines |
| Electric Machinery Fundamentals (5th Edition) |
| Electric Machinery |
| Electric Motors and Drives: Fundamentals Types and Applications |
| Electrical Machines and Drives: Fundamentals and Advanced Modelling |
| Electrical machines, drives, and power systems |

**Table 2:** Lecture sources which can be useful during the course of the lecture. For more information on sources, please consult the repo.

# Introduction i

| Topic | Units | Self Study |
|---|---|---|
| Magnetic Circuits and Materials | 4 | 4 |
| Transformers | 4 | 4 |
| Electromechanical Energy Conversion | 4 | 8 |
| Lab 1: Installation of a Drive System | 3 | 6 |
| Rotating Magnetic Fields | 2 | 4 |
| Lab 2: Building a Lathe | 4 | 8 |
| Synchronous Machines | 4 | 8 |
| Poly-phase Induction Drives | 4 | 8 |
| Lab 3: Synchronous and Induction Drives | 4 | 8 |

| Topic | Units | Self Study |
| --- | --- | --- |
| DC Drives | 4 | 8 |
| Special Machines | 2 | 3 |
| Lab 4: Photovoltaic Cells | 4 | 8 |
| Final Exam | 2 | 4 |
| Total | 45 | 90 |